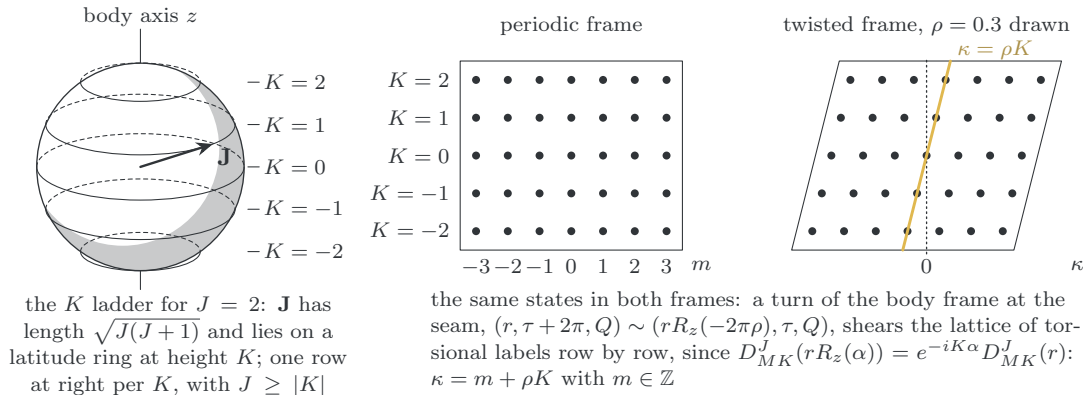
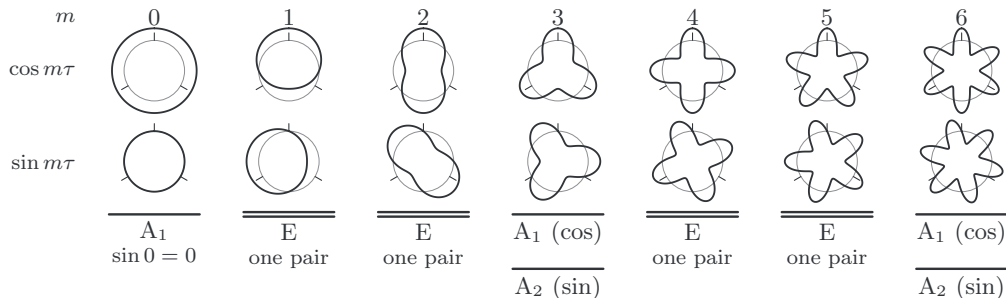


one state, two bases: $|r, \tau, Q\rangle_0$ and $|JMK, m, p\rangle$, related by the Wigner–Fourier transform on $SO(3) \times S^1 \times \mathbb{R}^d$ with the chart density absorbed, $\tilde{\psi} = \mathcal{J}^{1/2}\psi$; no energies are assigned



periodic scalar torsion on the circle of figure 9: $U_t f_m = e^{-2\pi i m/3} f_m$, $U_b f_m = f_{-m}$; the real modes drawn as radial plots, the three version marks at $\tau = 0, 2\pi/3, 4\pi/3$



level rules as in Harter (1993) fig. 4.2.3: a single rule for a one-dimensional species, a double rule for the two-dimensional E; the modes with $3 \mid m$ are visibly invariant under the third-turn t : $m = 0$ gives A_1 , and $m = 3, 6$ split into $A_1(\cos)$ and $A_2(\sin)$; every other (cos,sin) pair spans one E